Intelligent Age Verification

The Challenge: Verifing age and Preventing Fraud

Age-Gated Content

Verifying legal age requirements for restricted content and s ervices.

Anti-Bot Security

Distinguishing between real humans and automated systems attempting unauthorised access.

The Solution: A Dual-Layer Security System

Computer Vision

Precise age estimation from facial photographs using deep learning.

Speech Recognition

Real-time voice verification to confirm genuine user presence.

Computer Vision Module: Decoding Age from Faces

UTKFace Dataset

Over 20,000 human face images labelled with ages spanning 0 to 116 years for comprehensive training.

Pretrained CNN Architecture

Convolutional Neural Network pretrained on ImageNet, adapted through transfer learning for age estimation.

Training Configuration

MAE loss function, Adam optimiser (learning rate: 0.0001), batch size: 32, 20 epochs with validation checkpointing.

Genetic Optimisation: The Intelligence Behind the Model

Automated Parameter Tuning

Genetic Algorithmtreats hyperparameter combinations as population individuals, using validation MAE as fitness function.

Optimised Parameters

- Learning
- rate Batch
- size Dropout

  rate

4.93

Validation MAE achieved by best genetic individual

20K

Processed during training phase

32

Optimised parameter value

Voice Verification Module: The Anti-Fraud Barrier

01

Random Sentence Display

System presents unique phrase for user to read aloud.

02

Audio Capture

Microphone records user's spoken response.

03

Speech Recognition

Google Speech Recognition API transcribes audio to text.

04

Similarity Matching

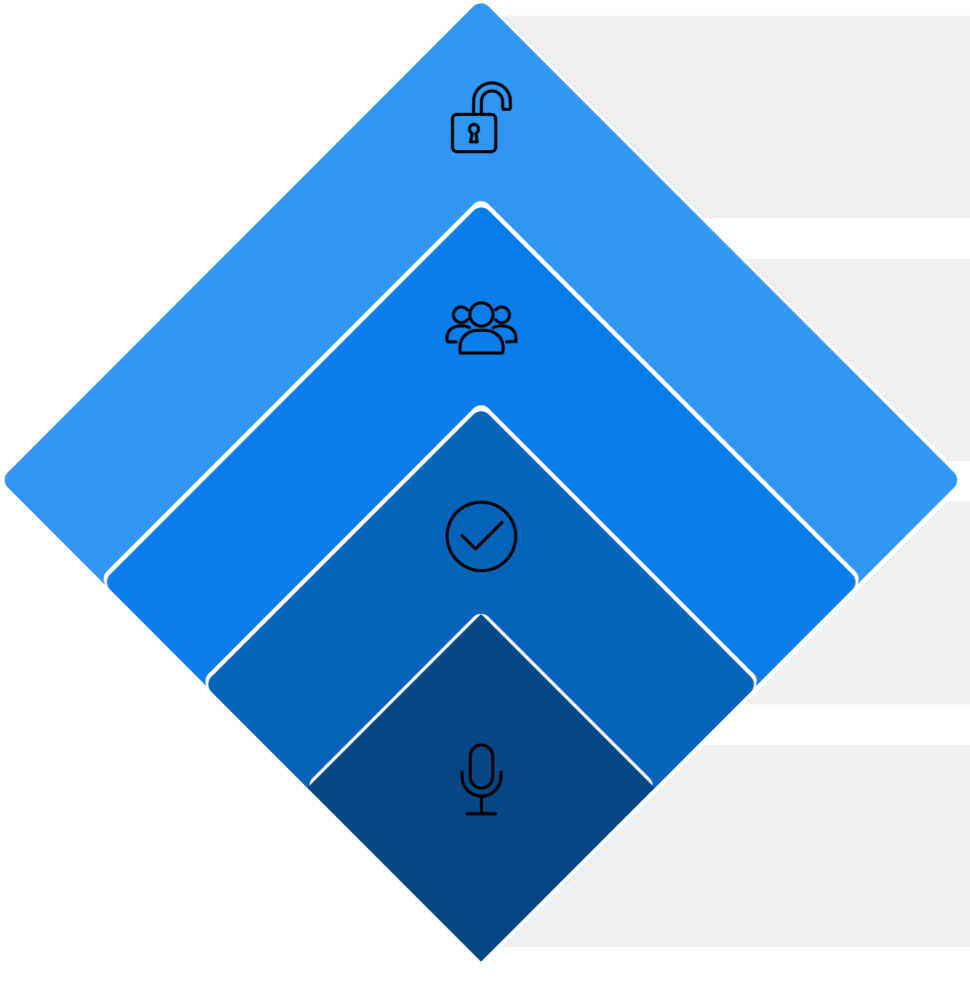
SequenceMatcher calculates similarity score against original s entence.

05

Threshold Validation

75%+ similarity grants access; lower scores prompt retry.

Verification Flow: Sequential Security



Access Granted

Allow or deny access

Estimate Age

Runageestimation model

Voice Success?

Decision: proceed or reject

Voice Check

Verify caller identity

Results and Limitations: A Realistic View

Successes

- Complete chained operation with local web
- interface Val MAE of 4.93 years on validation set
- Working voice verification under normal microphone conditions

Current Limitations

- Exactage prediction remains challenging
- Extended training times on laptop
- hardware Requires clear diction and quality microphones

The Future: Towards Age Groups and Greater Precision

Age Bracket Classification

Transition from exact age prediction to group classification (e.g., 0-17 minors vs 18+ adults).

Finer Age Ranges

Testing with smaller age group intervals to improve accuracy and reliability.